nutrients. After this period, the membrane is taken out and the colonies of bacterias are counted by means of microscope. This method is called membrane filter technique.

4.4 Treatment of Water

Objectives of water Treatment

The following are the objectives of water treatment

i. To remove the dissolved gases and colour from the water
ii. To remove the unpleasant and objectionable taste from the water
iii. To kill all the pathogens from the water
iv. To remove corrosive materials from the water
v. To make water fit for domestic and industrial purpose.

Flow Diagram of a Treatment plant **(Refer Page 72)**

Sequence of units

1. Intake point
2. Pump house
3. Plain Sedimentation tank
4. Coagulation tank
5. Filtration unit
6. Chlorination unit
7. Water softening plant
8. Overhead reservoir

4.4.1 Functions and locations of Treatment Units

The function of each unit is stated briefly

1. Intake point

The function of this unit is to collect water in the intake well so that the water can be supplied throughout the year.

2. Pump House

The function of this unit is to draw water from the intake point and to supply the same to the treatment plant.

3. Plain Sedimentation Tank

The function of this unit is to remove the heavier suspended particles in water. In this tank, the water is detained for some period or allowed to flow at a very low velocity so that heavier suspended particles are settled down at the bottom of sedimentation tank. But some lighter particles still remain in suspension.

4. Coagulation Tank

The function of this unit is to remove the lighter suspended particles by the application of some coagulants(chemicals). In this tank, some recommended coagulant is mixed with the water and the water is allowed to flow at a very low velocity through the coagulation tank. The coagulant makes the lighter particles to gain settleable size and ultimately settle down at the bottom of the tank. But some colloidal particles still remain in suspension.

5. Filtration unit

The function of this unit is to remove the finer colloidal particles and some bacteria by filtering media of sand and gravel. But some bacteria still remain in water.

6. Chlorination unit

The function of this unit is to destroy the bacteria by the application of chlorine.

7. Water softening tank

The function of this unit is to remove the hardness of water to make it

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